

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2022

TEST DATE: 13-02-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

Physics

PART – A

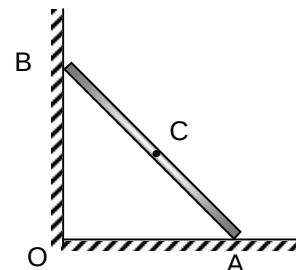
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

- The upper end of a simple pendulum is fixed to a vertical z-axis and set in motion such that the bob moves along a horizontal circular path of radius 1 m parallel to the x-y plane, 5 m above the origin. The bob has a speed of 4 m/s. The string breaks when the bob is vertically above the x-axis. Taking $g = 10 \text{ m/s}^2$, the point where it hits, the bob lies in the
 (A) x-y plane at (1 m, 4 m)
 (B) y-z plane at (1 m, 4 m)
 (C) x-y plane at (4 m, 1 m)
 (D) y-z plane at (5 m, 1 m)

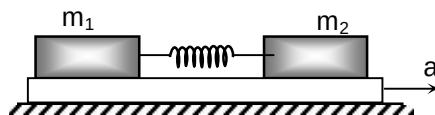
- Resistance of a conductor of length L and area of cross-section A is given by $R = \frac{\rho_0}{A} \left(L + \frac{1}{2} \alpha L^2 \right)$, where α is a constant and ρ_0 is resistivity at 0°C . Then the resistivity ρ of the conductor varies along the length x of the conductor as
 (A) $\rho = \rho_0(1 - \alpha x)$
 (B) $\rho = \rho_0(1 - \alpha x^2)$
 (C) $\rho = \rho_0(1 + \alpha^2 x)$
 (D) $\rho = \rho_0(1 + \alpha x)$

- A rod of length 2ℓ is moving such that its ends A and B move in contact with the horizontal floor and vertical wall respectively as shown in figure. O is the intersection point of the vertical wall and horizontal floor. The velocity vector of the centre of rod C is always directed along tangent drawn at C to the
 (A) circle of radius $\frac{\ell}{2}$ whose centre lies at O
 (B) circle of radius ℓ whose centre lies at O
 (C) circle of radius 2ℓ whose centre lies at O
 (D) None of these

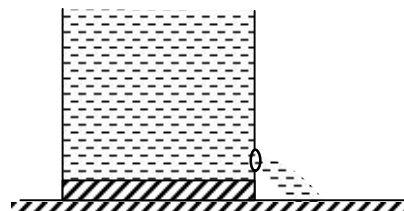


- The secondary coil of an ideal step down transformer is delivering 500 watt power at 12.5 A current. If the ratio of turns in the primary to the secondary is 5: 1, then the current flowing in the primary coil will be
 (A) 62.5 A
 (B) 2.5 A
 (C) 6 A
 (D) 0.4 A

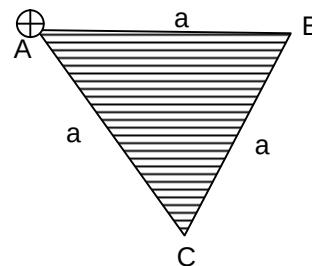
5. Two block of masses m_1 and m_2 are connected with a massless unstretched spring and placed over a plank moving with an acceleration 'a' as shown in figure. The coefficient of friction between the blocks and platform is μ .
- (A) spring will be stretched if $a \geq \mu g$
 (B) spring will be compressed if $a \leq \mu g$
 (C) spring will neither be compressed nor be stretched $a \leq \mu g$
 (D) spring will be in its natural length under all conditions



6. Liquid of density ρ is filled in the container of negligible mass having area A . Area of hole 'a' very small ($a \ll A$) is made near the bottom in the wall of container. Co-efficient of friction between container and surface is equal to $\left(\frac{a}{A}\right)$. As liquid is coming out of vessel, instantaneous acceleration of the container is:

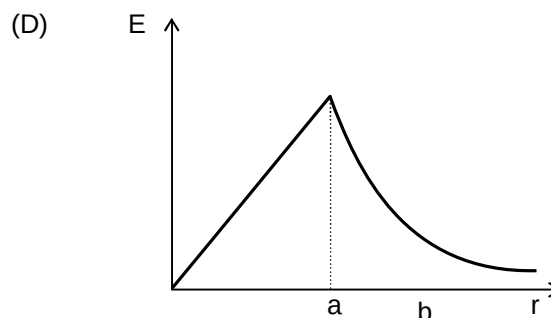
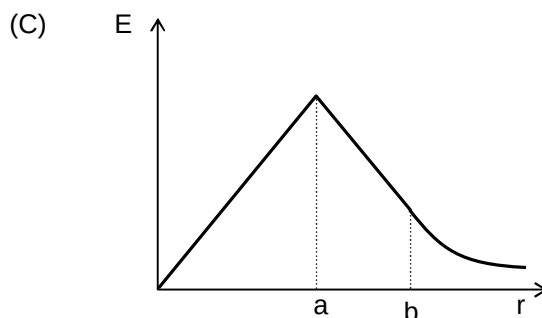
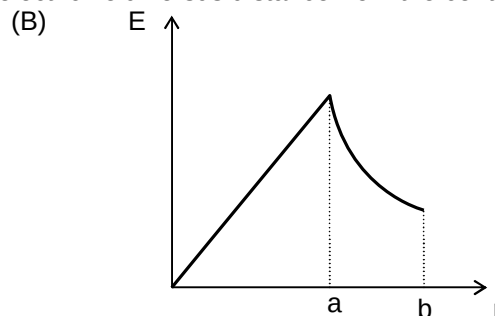
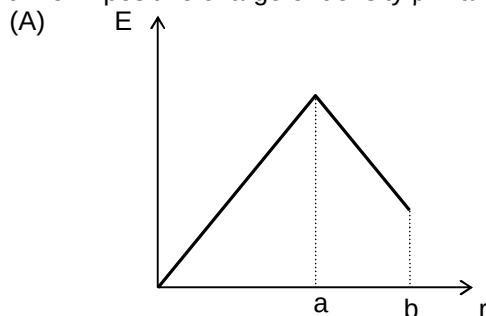


- (A) $\left(\frac{A}{a}\right)g$
 (B) $\frac{3g}{2}$
 (C) $\left(\frac{a}{A}\right)g$
 (D) zero
7. A uniform triangular plate ABC of mass m and moment of inertia I (about an axis passing through A and perpendicular to plane of the plate) can rotate freely in the vertical plane about point 'A' as shown in figure. The plate is released from the position shown in the figure. Line AB is horizontal. The acceleration of centre of mass just after the release of plate is:

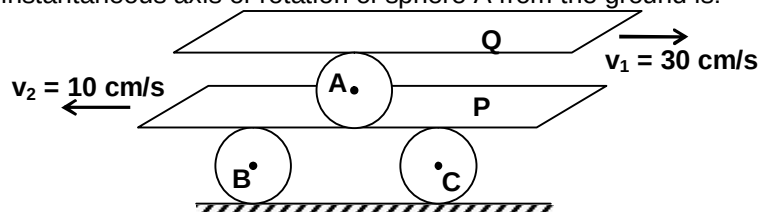


- (A) $\frac{mga^2}{\sqrt{3} I}$
 (B) $\frac{mga^2}{4 I}$
 (C) $\frac{mga^2}{2\sqrt{3} I}$
 (D) $\frac{mga^2}{3 I}$

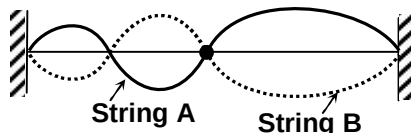
8. A capacitor is made of two concentric spherical shells A and B of radii a and b respectively, where $a < b$. The external shell B is grounded ($V_B = 0$). Inner shell i.e., ($r \leq a$) is filled with a uniform positive charge of density ρ . Variation of electric field versus distance from the centre.



9. All three spheres are identical having radius 10 cm. There is no slipping at any point of contact. Plank P and Q are moving in opposite direction with speed 10 and 30 cm/sec. height of instantaneous axis of rotation of sphere A from the ground is:

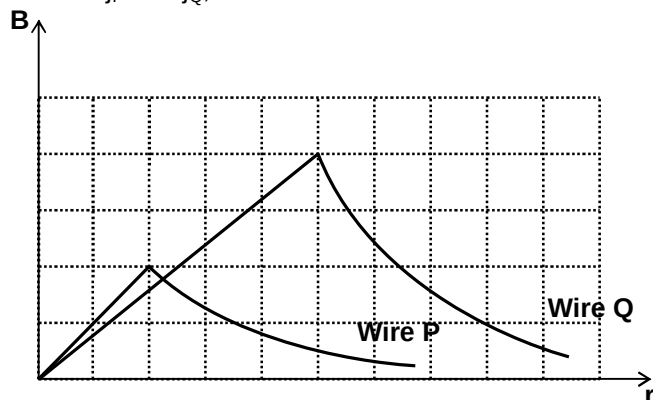


- (A) 20 cm
(B) 30 cm
(C) 15 cm
(D) 25 cm
10. Two strings A and B of equal length but unequal linear mass density are tied together with a knot and stretched between two supports. A particular frequency happens to produce a standing wave on each part, with node at the knot as shown. If linear mass density of the two strings A and B are μ_1 and μ_2 respectively, then:



- (A) $\mu_1 = \mu_2$
(B) $\mu_1 < \mu_2$
(C) $\mu_1 > \mu_2$
(D) can't be decided

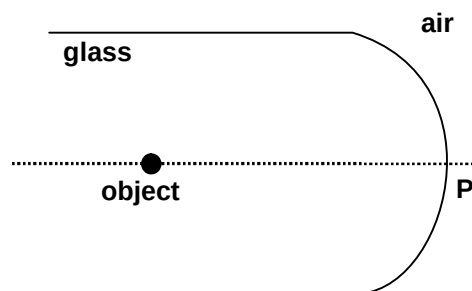
11. Figure shows variation of magnetic field inside and outside two cylindrical wires P and Q carrying currents that are uniformly distributed across the cross section of wires. If current density for two wires are j_P and j_Q , then:



- (A) $j_P = j_Q$
 (B) $j_P < j_Q$
 (C) $j_P > j_Q$
 (D) can't be decided

12. Radius of curvature of a concave spherical surface separating air-glass medium is R . A point object is placed on the principal axis in the glass ($\mu_g = 1.5$). For the image to be real, seen by observer in air, distance of the object from P should be:

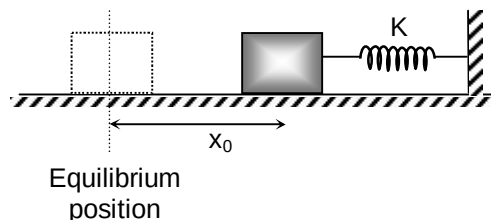
- (A) $2R < u < 3R$
 (B) $u < 2R$
 (C) $u = R$
 (D) $u > 3R$



13. A parent radioactive nucleus A (decay constant λ_A) converts into a radio-active nucleus B of decay constant λ_B , initially number of atoms of B is zero. At any time N_A , N_B is number of atoms of nuclei A and B respectively then maximum value of N_B is:

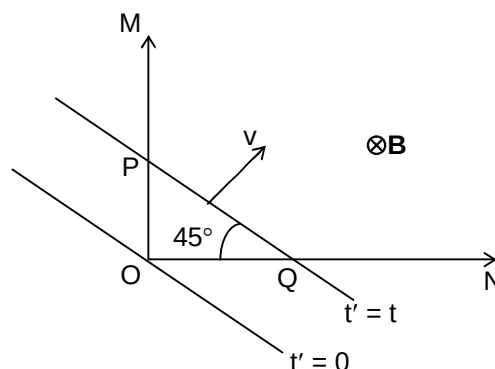
- (A) $\frac{\lambda_A N_A}{\lambda_B}$
 (B) $\frac{\lambda_B N_A}{\lambda_A}$
 (C) $\frac{(\lambda_A + \lambda_B) N_A}{\lambda_B}$
 (D) $\frac{\lambda_B N_A}{(\lambda_A + \lambda_B)}$

14. Initially spring is compressed by distance x_0 from equilibrium position. At this compression block is given velocity $\sqrt{\frac{3K}{m}}x_0$, so that compression in the spring increases and block start S.H.M. (Spring constant is K). Equation of motion of the block is:



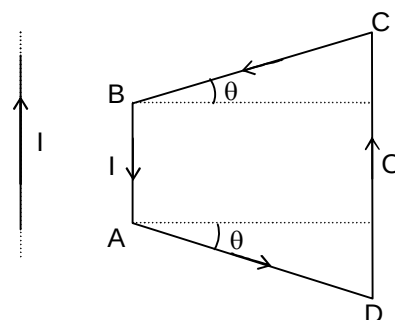
- (A) $y = 3x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{3}\right)$
 (B) $y = 2x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{6}\right)$
 (C) $y = 3x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{6}\right)$
 (D) $y = 2x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{3}\right)$

15. An electric circuit consists of three conducting rods MO, ON and PQ as shown in the figure. Resistance of rods per unit length is λ . The rod PQ slides at a constant velocity v , keeping its tilt angle relative to ON and MO fixed at 45° . At each instant, circuit is closed. The system is placed in a uniform magnetic field B which is directly perpendicular to the page. The rate of change of induced emf at any instant of time t is



- (A) Bv
 (B) Bv^2
 (C) Bv^3
 (D) B^2v
16. 1.5 mW of 400 nm light is directed at a photoelectric cell. If 0.10% of the incident photons photoelectrons, the current in the cell is
- (A) $0.36 \mu\text{A}$
 (B) $0.48 \mu\text{A}$
 (C) 0.42 mA
 (D) 0.32 mA

17. Loop ABCD and long straight wire are in the same plane. Side $CD = 2AB$. Distance between long straight wire and AB is equal to the distance between AB and CD. Net force on the loop is:



- (A) Towards the long straight wire
 (B) Away from the long straight wire
 (C) Parallel to the long straight wire
 (D) Net force on the loop will be zero

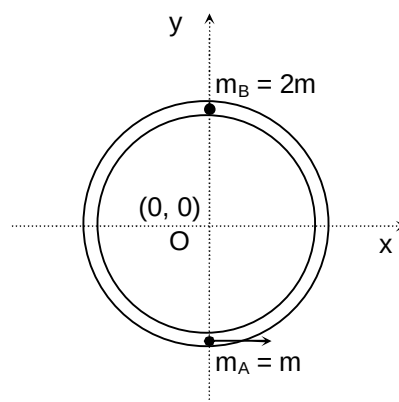
18. A hollow circular tube of radius R and negligible internal diameter is fixed on horizontal surface. Ball A of mass m is given velocity v_0 in the shown direction. It collides with ball B of mass $2m$. Collision is perfectly elastic. If centre of loop is origin of co-ordinate system, then co-ordinate of next collision.

(A) $\left(\frac{\sqrt{3}}{2}R, -\frac{R}{2}\right)$

(B) $\left(-\frac{\sqrt{3}}{2}R, \frac{R}{2}\right)$

(C) $\left(\frac{R}{2}, \frac{\sqrt{3}}{2}R\right)$

(D) $\left(\frac{R}{2}, \frac{R}{2}\right)$



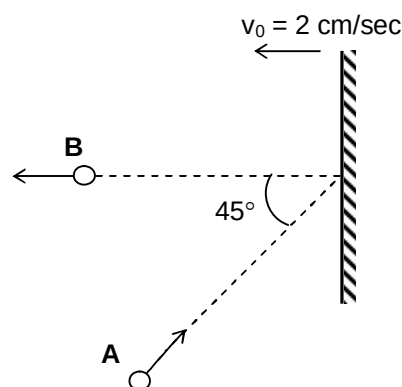
19. Mirror is moving towards the particles with speed 2 cm/sec . Speed of particles A and B are $10\sqrt{2}$ and 5 cm/sec respectively in the direction shown in figure. Magnitude of velocity of image of the particle B with respect to image of A.

(A) $\sqrt{325} \text{ cm/sec}$

(B) 15 cm/sec

(C) 13 cm/sec

(D) $\sqrt{269} \text{ cm/sec}$



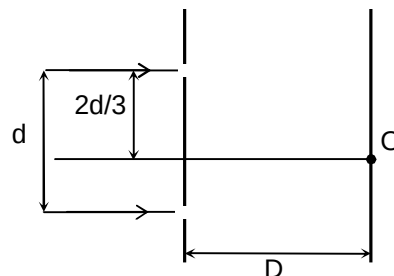
20. In the figure shown if a parallel beam of white light is incident on the plane of the slits, then the distance of central white spot on the screen from point O is: [assume $d \ll D$, $\lambda \ll d$]

(A) 0

(B) $d/2$

(C) $d/3$

(D) $d/6$

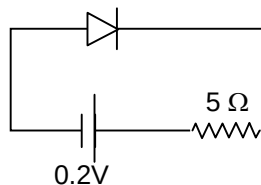


SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. The least count of a stopwatch is $\frac{1}{5}$ second. The time of 20 oscillations of a pendulum is measured to be 50 sec. What will be the minimum percentage error in the measurement of time.

22. Determine the current in the circuit in ampere.



23. A vernier calipers has 1 mm marks on the main scale. It has 10 equal divisions on the vernier scale which match with 8 main scale divisions. What will be the least count of vernier calipers in cm.

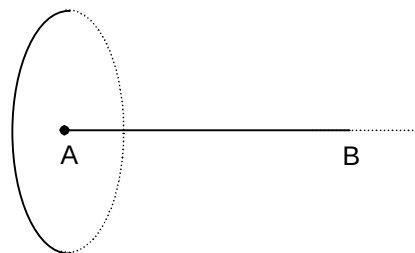
24. The angle of dip at a place where horizontal and vertical components of earth's magnetic field are equal is $\frac{\pi}{N}$. Find the value of N.

25. A sinusoidal electromagnetic plane wave travels through empty space in x-direction. At a certain point P in space, the oscillating electric field E_y has amplitude of magnitude $E_0 = 63$ V/m. Find the maximum magnitude B_0 of the magnetic field at the same point in microtesla. (speed of light $c = 3 \times 10^8$ m/s)

26. Three point charges, each of charge $+q$ are rigidly fixed at the three corners of an equilateral triangle of side a . The surface tension of soap film is T . The system of charges and plane film are in equilibrium and $a = k \left(\frac{q^2}{T} \right)^{1/N}$, where k is constant. Then find N.

27. n moles of an ideal monoatomic gas undergoes a process in which the temperature changes with volume as $T = kV^2$. If the temperature changes from T_0 to $4T_0$ then the work done by the gas is $\frac{N}{2} nRT_0$. Find the value of N.

28. A semi-infinite conductor AB lies along the axis of conducting circular loop of radius R , such that one end of conductor is at the centre of loop. If the current in conductor AB varies at a rate of 0.2 amp/sec then what will be induced emf (in volt) in the loop. (Radius of loop $R = 2$ m)



29. A complex current wave is given by $i = 3 + 3 \sin \pi t$. What will be its average value over one time period in ampere?

30. A charge $+Q$ is uniformly distributed in a spherical volume of radius R . A particle of charge $+q$ and mass m is projected with velocity v_0 from the surface of spherical volume towards its centre. The minimum value of v_0 such that it just reaches at a distance $\frac{R}{2}$ from the centre of spherical volume is $\sqrt{\frac{3Qq}{Nm\pi\epsilon_0 R}}$. Find the value of N. (There is no other resistance force on charged particle except electric force)

Chemistry

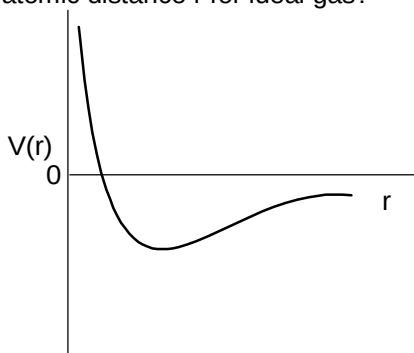
PART – B

SECTION – A (One Options Correct Type)

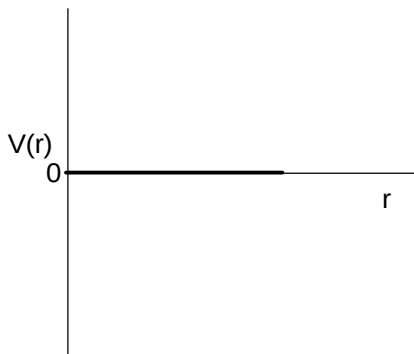
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following graphs represents the relation between inter atomic potential energy $V(r)$ and inter atomic distance r for ideal gas?

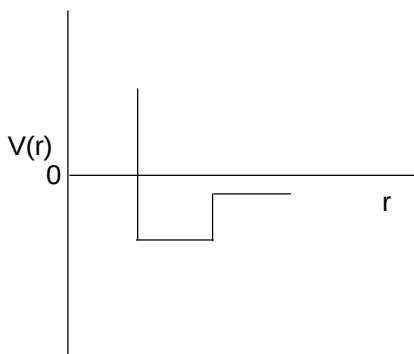
(A)

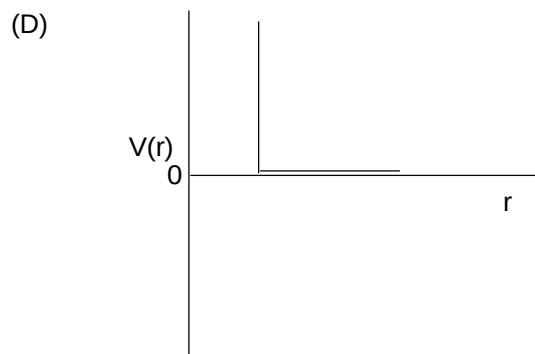


(B)



(C)

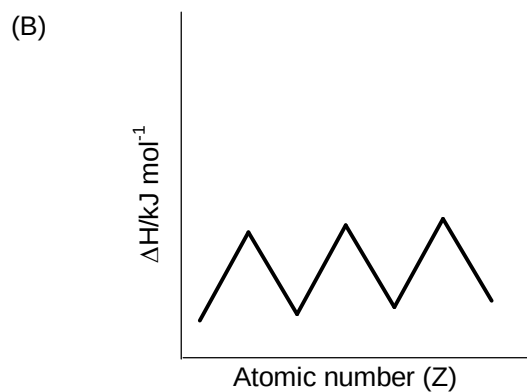
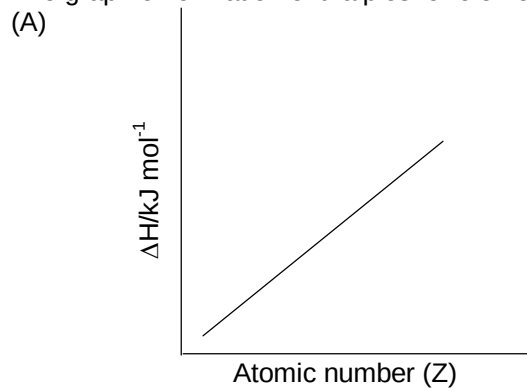


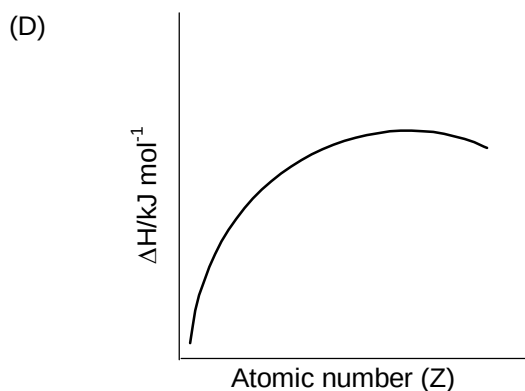
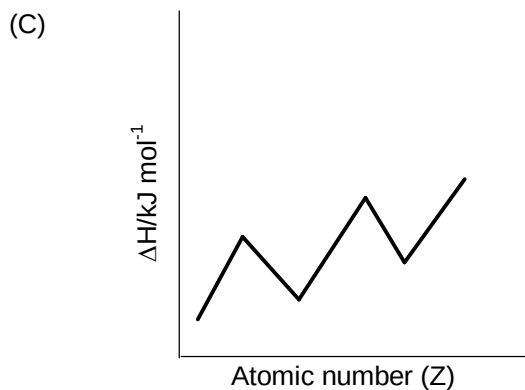


32. In which of the following molecules, the shortest O – O bond length can be expected?

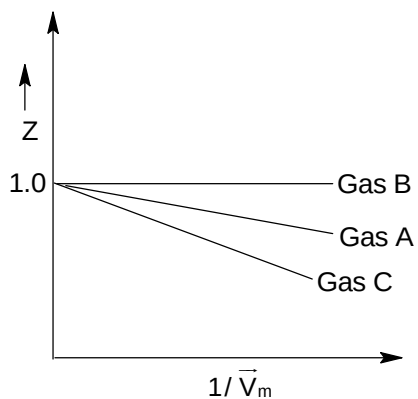
- (A) O_3
- (B) O_2
- (C) O_2F_2
- (D) KO_2

33. The graph of ionization enthalpies for elements with atomic number 3 to 10 is best shown as





34. Considering the graph, which of the following gases has the highest critical temperature T_c ?

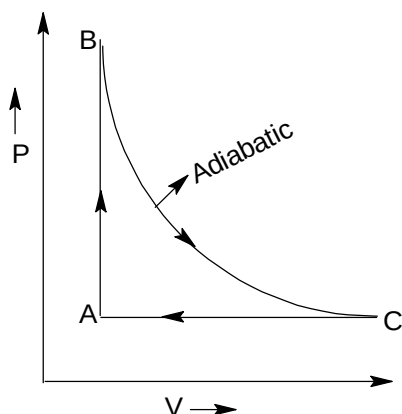


- (A) A
(B) B
(C) C
(D) All have identical T_c
35. Which of the following alkali metal is not miscible with other alkali metals in their molten phase?
- (A) Li
(B) K
(C) Rb
(D) Cs

36. Which of the following is correct reason behind greater volatility of reaction between potassium with water compare to that of lithium?
- K has greater reducing nature than Li
 - K has lower reducing nature than Li
 - K has high melting point than Li
 - K has low melting point than Li

37. One mole of an ideal mono-atomic gas is carried around the cycle as shown in the figure below. Find the amount of heat added in the process A to B is

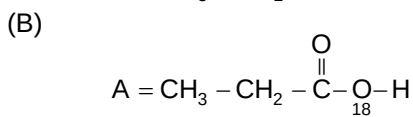
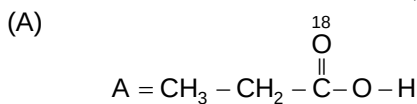
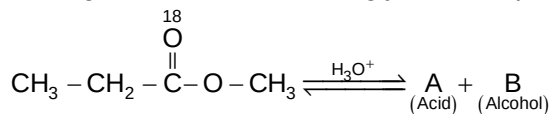
$$\left(\text{Given : } C_V = \frac{3R}{2} \right)$$



(Given: $T_A = 300 \text{ K}$, $T_B = 600 \text{ K}$, $T_C = 455 \text{ K}$)

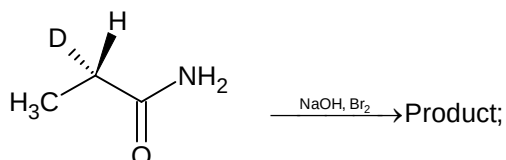
- 3742 J
 - 3742 J
 - 2812 J
 - 3228 J
38. Which of the following statement is not correct?
- With increase in temperature, the fraction of molecules having energy lower than activation energy decreases.
 - With increase in temperature, the collision frequency of reactant molecules increases.
 - With decrease in temperature, the pre-exponential factor in collision theory increases.
 - Rate constant can be less than pre-exponential factor in Arrhenius equation.
39. Find the incorrect statement.
- The concentration of oxygen in air is greater than the concentration of oxygen in Lake's water.
 - In cold winter season, thick ice forms on the surface of water in Lake; this leads to restriction on light entering into Lake and hence photosynthesis of plants under water stops. This consequently leads to increase in concentration of oxygen in the water
 - The irritant red haze in the traffic and congested places is due to oxides of nitrogen
 - Carbon dioxide gas is confined to troposphere only
40. The addition of NaH_2PO_4 to $0.1 \text{ M H}_3\text{PO}_4$ will cause
- no change in pH value
 - increase in its pH value
 - decrease in its pH value
 - change in pH but cannot be predicted

41. If experimentally, it has been found that there are two types of C – C bond length in cyclobuta-1,3-diene. Based on this, the compound will be
 (A) aromatic
 (B) anti-aromatic
 (C) non-aromatic
 (D) can not predict
42. The total number of paramagnetic compounds in the following group is
 $\text{Na}_3[\text{CoF}_6], \text{Na}_2\text{O}_2, \text{CsO}_2, \text{K}_3[\text{Co}(\text{CN})_6], [\text{V}(\text{H}_2\text{O})_6]_3, [\text{FeF}_6]_2, \text{K}_3[\text{Fe}(\text{H}_2\text{O})_6], \text{Ni}(\text{CO})_4$
 (A) 4
 (B) 5
 (C) 3
 (D) 6
43. In Ellingham diagram, where we plot ΔG° for the formation of oxides of various metal with respect to temperature. Which of the following statement (s) is/are correct?
 I. A metal with higher slope of curve gets always oxidized.
 II. A metal with lower slope of curve never get reduced at any temperature.
 III. The slope of the curve for some metal changes after particular temperature because of phase change of metal.
 IV. If two metals have same slopes, the one which has greater y-intercept usually get oxidized by the other one.
 V. If two metals have same slopes, the one which has greater y-intercept usually get reduced by the other one.
 (A) II, III and V
 (B) III and V
 (C) All are wrong
 (D) III, IV and V
44. In the series Sc (Z = 21) to Zn (Z = 30), the enthalpy of atomization is lowest for which metal?
 (A) Sc
 (B) Mn
 (C) Cu
 (D) Zn
45. How many P = O bonds are present in Orthophosphoric acid?
 (A) 1
 (B) 2
 (C) 0
 (D) 3
46. In the given reaction, the wrongly matched product is



- (C) $B = \text{CH}_3 - \overset{18}{\text{O}} - \text{H}$
 (D) $B = \text{CH}_3 - \text{O} - \text{H}$

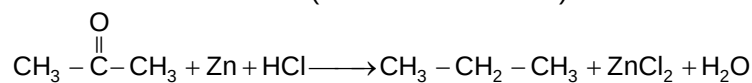
47.



The possible product is:

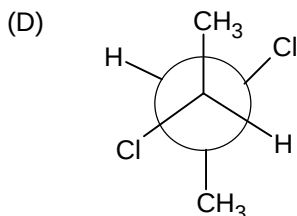
- (A)
- (B)
- (C)
- (D)

48. Clemmensen's reduction (unbalanced reaction)



Find the stoichiometric coefficient of zinc in the balanced Clemmensen's reduction reaction if stoichiometric coefficient of acetone is 2.

- (A) 4
 (B) 2
 (C) 8
 (D) 6
49. Which of the following compound can produce chiral compound upon treatment with CH_3MgCl and then acidification?
 (A) Formaldehyde
 (B) Pentanone
 (C) 2-hexanone
 (D) 3-hexanone
50. Which of the following compound is chiral?
 (A) $\text{CH}_3 - \underset{(\text{cis})}{\text{CH}} = \underset{\text{R}}{\text{CH}} - \underset{(\text{cis})}{\text{CH}} = \text{CH} - \text{CH}_3$
 (B) $\text{CH}_3 - \underset{(\text{cis})}{\text{CH}} = \underset{\text{S}}{\text{CH}} - \text{CH} = \underset{(\text{trans})}{\text{CH}} - \text{CH}_3$



58. A graph is plotted between $\log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$ (Y – axis) and E_{cell} (x – axis). The curve is linear with intercept on y-axis equal to 37.28. Calculate E_{cell} for the cell
 $\text{Zn(s)} | \text{Zn}^{2+} (0.1 \text{ M}) || \text{Cu}^{2+} (0.01 \text{ M}) | \text{Cu}$
59. A solution containing equimolar mixture of lead acetate and silver nitrate is titrated with H_2S solution. And, then the solution is boiled with hot water. After filtration, the weight of precipitate was 6.2 gm. Assuming the yield of the experiment to be 100%, find the mass of lead acetate that was present in the initial mixture (in gm). [Given: Atomic mass, Pb = 207.3, Ag = 108, S = 32].
60. A 100 mL of 0.1 M metallic-copper colloidal solution is titrated with sufficient amount of dilute nitric acid. Find the mass (in gm) of the oxide of nitrogen which has been formed.

Mathematics**PART – C****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let A and B are two non-singular matrices of order 3 such that $A + B = 2I$ and $A^{-1} + B^{-1} = 3I$, then AB is equal to (where I is the identity matrix of order 3)
- (A) $2I$
(B) $\frac{2I}{3}$
(C) A
(D) B
62. If $\lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \ln(1+x) - 2x^3 + x^4}$ exists and finite, then which of the following is not correct?
- (A) $a = 6$
(B) $a = b$
(C) $c = 2$
(D) $c = 0$
63. If $A = \begin{bmatrix} 5 & 2016 \\ 1 & 403 \end{bmatrix}$, then the value of $|A^{2015} - 5A^{2014}|$ is
- (A) 1
(B) -1
(C) 5
(D) -2016
64. If $\vec{a}$ and $\vec{b}$ are two non-collinear unit vectors such that $|\vec{a} \times \vec{b}| = \left| \frac{\vec{a} - \vec{b}}{2} \right|$, then the value of $|\vec{a} \times (\vec{a} \times \vec{b})|^2$ is
- (A) $\frac{3}{4}$
(B) $\frac{2}{3}$
(C) $\frac{1}{2}$
(D) $\frac{1}{4}$
65. $\int (x^{24} + x^{16} + x^8)(2x^{16} + 3x^8 + 6)^{\frac{1}{8}} dx = \frac{(2x^{24} + 3x^{16} + 6x^8)^{\frac{\beta}{\gamma}}}{\alpha} + c$
(where β, γ are co-prime natural numbers and c be constant of integration), then the value of $\alpha + \beta + \gamma$ is
- (A) 61
(B) 51
(C) 71
(D) 91

66. PQR is a triangular park with $PQ = PR = 200$ m. A tower stands at the mid-points of QR. If the angles of elevation of the top of the tower at P, Q and R are respectively 45° , 30° and 30° , then the height of the tower (in m) is
 (A) 100
 (B) 200
 (C) 150
 (D) 125
67. Let $L_1 : x = y = z$ and $L_2 : x - 1 = y - 2 = z - 3$ be two lines. The foot of perpendicular drawn from the origin $O(0, 0, 0)$ on L_1 to L_2 is A. If the equation of a plane containing the line L_1 and perpendicular to OA is $10x + by + cz = d$, then the value of $b + c + d$ is
 (A) 12
 (B) -12
 (C) 10
 (D) -10
68. If $y = \tan^{-1}\left(\frac{4x}{5x^2 + 1}\right) + \tan^{-1}\left(\frac{2 + 3x}{3 - 2x}\right)$ where $x \in \left(0, \frac{2}{3}\right)$. If $\frac{dy}{dx} = \frac{\lambda}{1 + kx^2}$, then $\left|\frac{\lambda + k}{10}\right|$ is
 (where λ and k are natural numbers)
 (A) 5
 (B) 3
 (C) 2
 (D) 6
69. Let $R = \{(a, b) : a^2 + b^2 = 4, a, b \in \mathbb{R}\}$, then relation R is
 (A) symmetric
 (B) transitive
 (C) reflexive
 (D) anti-symmetric
70. The contrapositive of the conditional "If Ram works hard, then he gets a first class" is
 (A) if Ram does not work hard, then he does not get a first class
 (B) if Ram does not get a first class, then he works hard
 (C) if Ram does not get a first class, then he does not work hard
 (D) if Ram get a first class, then he does not work hard
71. Number of complex numbers satisfying the relation $|z + \bar{z}| + |z - \bar{z}| = 2$ and $|z + i| + |z - i| = 2$, is
 (A) 1
 (B) 2
 (C) 3
 (D) 4
72. The coefficient of x^{13} in the expansion of $(1 - x)^5(1 + x + x^2 + x^3)^4$ is
 (A) 24
 (B) 12
 (C) 6
 (D) 4

73. If $4\cos 36^\circ + \cot 7\frac{1}{2}^\circ = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3} + \sqrt{n_4} + \sqrt{n_5} + \sqrt{n_6}$ (where $n_1, n_2, n_3, n_4, n_5, n_6$ are natural numbers), then the value of $\sum_{i=1}^6 (n_i)^2$
- (A) 19
(B) 91
(C) 81
(D) 101
74. The value of $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{3n^2 - 3n + 1}{n^6 - 3n^5 + 3n^4 - n^3 + 1} \right)$ is
- (A) $\frac{\pi}{2}$
(B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$
(D) $\frac{\pi}{6}$
75. If a variable takes the values 0, 1, 2,, n with frequencies $1, {}^nC_1, {}^nC_2, \dots, {}^nC_n$, then the Arithmetic Mean is
- (A) $\frac{2^n}{n}$
(B) $\frac{n}{2}$
(C) $n + 1$
(D) n
76. If $f(\theta) = |\cos \theta + 1| + |\cos \theta - 2| + |\cos \theta - 1| + |\cos \theta - 3|$, where $0 \leq \theta \leq 2\pi$, then the product of maximum and minimum values of x is
- (A) 45
(B) 30
(C) 27
(D) 36
77. The real values of λ such that $(\lambda^2 - 4\lambda + 3)x^2 + (\lambda^2 - 5\lambda + 6)x + (\lambda^2 - 9) = 0$ has more than two roots is
- (A) 2
(B) 1
(C) 0
(D) 3
78. If $\frac{1}{x^2} + \frac{1}{y^2} = 1$, then the maximum value of $\frac{5x + 12y + 7xy}{xy}$ is
- (A) 7
(B) 13
(C) 15
(D) 20

79. If $\sum_{n=1}^{360} \left(\frac{1}{n\sqrt{n+1} + (n+1)\sqrt{n}} \right) = \frac{p}{q}$ (where p and q relatively prime integers), then $|p - q|$ is
 (A) 0
 (B) 1
 (C) 2
 (D) 3
80. If $f^{-1}(x) = x^3 + 3x - 4$, then the value of $|36f''(0)|$ is
 (A) 2
 (B) 3
 (C) 1
 (D) 0

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Let $f(1) + g(1) = 9e$; $f(x) = -x^2 g'(x)$; $g(x) = -x^2 f'(x)$. If $\int_1^4 \left(\frac{f(x) + g(x)}{x^2} \right) dx = k \left(e - e^{\frac{1}{4}} \right)$, then $\left| \frac{k}{2} \right|$ is
82. If $y = a \cos(e^x) + b \sin(e^x)$, then $\frac{y'' + ye^{2x}}{y'}$ is
83. In the coordinate plane, ellipse $C_1 : \frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1$ ($a_1 > 0, b_1 > 0$) and hyperbola $C_2 : \frac{x^2}{a_2^2} - \frac{y^2}{b_2^2} = 1$ ($a_2, b_2 > 0$) has the same focus points F_1, F_2 . Point P is one of the intersection points of C_1 and C_2 and PF_1 is perpendicular to PF_2 . If e_1 is the eccentricity of C_1 and e_2 is the eccentricity of C_2 . Then minimum value of $9e_1^2 + e_2^2$ is
84. If x_1, x_2, x_3, x_4 and x_5 be 5 positive numbers such that $x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 15$ and $x_1 \cdot x_2^2 \cdot x_3^3 \cdot x_4^4 \cdot x_5^5 = 1$, then $\frac{x_1 + x_2 + x_3 + x_4 + x_5}{2}$ is
85. If tangent drawn to the parabola $y^2 = -ax$ ($a > 0$) from a point A(1, 0) which also touches the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ at B such that $\angle ASB = 90^\circ$, where S is the focus of the hyperbola, then the value of $a + b^2$ is
86. Let E_1, E_2, E_3 be three independent events associated with a random experiment such that $3P(E_1 \cap \bar{E}_2 \cap \bar{E}_3) = P(\bar{E}_1 \cap E_2 \cap \bar{E}_3) = 9P(\bar{E}_1 \cap \bar{E}_2 \cap E_3) = 3 - 3P(E_1 \cup E_2 \cup E_3)$, where $P(E_1), P(E_2), P(E_3) \neq 1$ and $P(A)$ denotes probability of event A. If absolute value of $\begin{vmatrix} P(E_1) & P(E_2) & P(E_3) \\ P(E_2) & P(E_3) & P(E_1) \\ P(E_3) & P(E_1) & P(E_2) \end{vmatrix} = \frac{a}{b}$, where $a, b \in \mathbb{N}$, then the least value of $\left(\frac{a+b}{10} \right)$ is

87. A number x is selected from the set of first 9 natural numbers (i.e., $x = 1, 2, 3, \dots, 9$). If the probability that $f(f(x)) = x$ where $f(x) = x^2 - 3x + 3$ is $\frac{m}{9}$, then m is equal to
88. If $\alpha, \beta, \gamma, \delta$ be four roots of the equation $x^4 - 10x^3 + 37x^2 - 60x + 32 = 0$.
If $\frac{1}{\alpha^2 - 5\alpha + 10} + \frac{1}{\beta^2 - 5\beta + 10} + \frac{1}{\gamma^2 - 5\gamma + 10} + \frac{1}{\delta^2 - 5\delta + 10} = \frac{p}{q}$, where p and q are co-prime numbers, then $p + q$ is
89. A straight line L with the slope $\frac{3}{4}$ touches a circle whose radius is 5 units and centre lies on the x -axis. If the length of this tangents from a point on the x -axis is $\frac{a}{b}$ where $a, b \in \mathbb{N}$, then the least values of $\frac{a+b}{10}$ is
90. The coefficient of x^8 in the expansion of $(2 + 9x^3 + 6x^4 + x^5)^{10}$ is $5 \cdot 2^p \cdot 3^q$, where $p, q \in \mathbb{N}$, then $\frac{p+q}{5}$ is